The Fool's Theorem

Error Gravitation in Long-Horizon Agents:
Attention Dilution, the Dilution Threshold, and Bounded Context as a Remedy

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Abstract

We give a complete account of a failure mode in long-horizon tool-using agents that we call *Error Gravitation*, and that we refer to informally as the *Fool's Theorem*. The name is a label for the phenomenon, not a claim of mathematical novelty, and §1 states that distinction precisely.

The account has three parts. First, we formalize the mechanism. Given an attention query whose keys split into a fixed invariant set of size L_u and a growing distractor set of size N, the invariant share satisfies an exact identity

$$A_{i,\mathrm{inv}} = \sigma\left(\Delta + \ln \frac{L_u}{N}\right),$$

where Δ is the mean logit gap separating the two sets and σ is the logistic function. Two consequences follow immediately and are sharp rather than worst-case: $A_{i,\mathrm{inv}} > \frac{1}{2}$ if and only if $N < L_u e^{\Delta}$, and bounding $N \le C$ yields the closed-form guarantee $A_{i,\mathrm{inv}} \ge \big(1 + (C/L_u)e^{-\Delta}\big)^{-1}$, which requires no norm-boundedness assumption and is non-vacuous at realistic scale. The logarithm in the threshold is the entire story: a model must supply a logit gap growing as $\ln N$ merely to hold the invariant share constant.

Second, we measure this on a real transformer. Using Qwen2.5-0.5B in eager mode and reading the true final-query attention distribution, we find that the effective gap follows

$$\hat{\Delta}(N) \;\approx\; 0.945 \ln N - 4.195 \qquad (R^2 = 0.9996),$$

a slope indistinguishable from 1. The model therefore compensates for context growth at almost exactly the rate the theory says is necessary, and the observed dilution is far weaker than the uniform-score analysis predicts: the invariant share falls only from 0.427 to 0.388 across a 32× increase in distractor volume. We regard this as the single most important finding in the paper, and it tempers our own remedy: on this model, with this distractor distribution, the invariant share does not collapse. What volume control buys is not rescue from collapse but a bound that does not depend on the model happening to compensate.

Third, we describe and prove the systems response. *Generational Context* tiers active context by invariance and places a hard ceiling on the mutable tiers; *Attention-Weighted Eviction* maintains that ceiling as an enforced post-condition rather than an assumption. Under it, the closed-form guarantee becomes horizon-independent. We prove the resulting *Generational Retention Guarantee*, and we state a set of falsifiable predictions that would refute the theory, together with the experiments that would test them.

The paper is deliberately two-sided. It argues that volume is the variable a system designer can control, and it documents the evidence that the model was already partly controlling it on its own.

1 On the name, and what is being claimed

The phrase *Fool's Theorem*, as used in this project, names a *phenomenon*: the tendency of a long-horizon agent to abandon its stated objective and settle into a loop of local, symptomatic repairs. It is not a theorem in the mathematical sense, and nothing in this document should be read as claiming otherwise.

This distinction is not pedantry. The failure mode is real and worth naming. But the mathematics that explains it is elementary, and the two results this paper actually proves are named separately:

- the **Softmax Inundation Bound** (§5), a worst-case statement derived from bounded projection norms; and
- the **Dilution Threshold** (§4), an exact identity relating the invariant attention share to the mean logit gap; and
- the **Generational Retention Guarantee** (§9), a horizon-independent lower bound under a maintained volume ceiling.

The third is the result we would defend under review. The second we consider the useful one in practice. The first is included because it is the statement most likely to be quoted, and it is the weakest of the three, so we would rather present it with its limitations attached than have it circulate without them.

> **A note on how to read the empirical section.** The measurements in §6 do not confirm the remedy. They partly undercut the motivation for it, by showing that the model we tested compensates for context growth far better than a naive analysis predicts. We report this prominently because a paper that only reported the favourable half of its own evidence would not be worth the reader's time.

2 The failure mode

2.1 What it looks like

Long-horizon agents fail in a characteristic way that is distinct from, and easier to mistake for, ordinary reasoning error. The symptoms are consistent across domains:

(a) **Objective drift.** By the middle of a session, the agent is optimizing something adjacent to, but not identical with, the stated goal. It fixes the error it most recently saw rather than the error that caused the run.

(b) **Symptom loops.** Successive turns apply variations of the same local repair. The agent reports progress because the immediate observation changed.

(c) **Recency dominance.** The most recent tool output determines the next action, regardless of whether it is diagnostically relevant. A misleading error message encountered late in a run outranks a decisive one encountered early.

(d) **Loss of constraint.** Constraints stated at the start ("do not restart the service", "do not modify the schema") stop constraining behaviour, not because the agent decided to violate them but because they have stopped competing for attention.

None of these require the model to be incapable. An agent can be entirely competent at local reasoning and still exhibit all four, because the failure is in *what the model is looking at*, not in how well it thinks about what it looks at.

2.2 A worked trace

The following is a condensed illustration, using a production-incident shape that recurs in practice. Only the parts that matter are shown.

```
turn 1  USER: Build is failing after deploy.  Find the root cause.  Do not
       restart services.  Do not modify the database schema.
turn 1  ASSISTANT: I'll start by checking the deploy logs.
turn 2  TOOL: ERROR connection refused to redis:6379
turn 2  ASSISTANT: A Redis connectivity failure.  Restarting the cache.
turn 3  TOOL: ERROR connection refused to redis:6379
turn 3  ASSISTANT: The restart did not help.  Restarting the cache again.
turn 4  TOOL: WARN migration 0042 not applied; column user.locale missing
turn 4  ASSISTANT: Redis is still unreachable.  Increasing the retry budget.
turn 5  TOOL: ERROR connection refused to redis:6379
turn 5  ASSISTANT: The cache is the blocker.  Restarting it once more.
turn 6  TOOL: ERROR schema mismatch on user.locale
turn 6  ASSISTANT: Restarting the cache to clear the schema error.
```

Turn 4 contains the root cause and turn 6 repeats it, but neither displaces the Redis failure from the agent's working set. The agent also violates two explicit constraints from turn 1. It is not that the model cannot read; it is that the constraint tokens and the diagnostic tokens have both stopped winning attention against a growing body of identical error lines.

Note what makes this example instructive: the distractor is not merely noise. It is a *plausible* error, which is why it holds attention so effectively. The failure mode is worst when the noise is superficially actionable.

2.3 Why the mechanism is not the obvious one

The natural diagnosis is capacity: the context is full, so the model has forgotten the goal. That diagnosis is usually wrong, or at least incomplete. Modern windows are large enough that a trace like the one above occupies a tiny fraction of them, and the instruction is still present in the input. What has changed is not availability but *allocation*: the instruction is present and receives a negligible share of attention mass.

This is the distinction the paper formalizes. It is also why the fix is not "use a bigger window": doubling the window doubles the room for distractors as well.

3 Formal setting

3.1 Transcript and length

Let an agent interact with an environment over discrete turns $t \in \mathbb{N}$. Define the initial context length

$$N_t = L_u + \sum_{k<t} \big(L_a^{(k)} + L_o^{(k)} + L_r^{(k)} \big), \tag{1}$$

where L_u is the length of the invariant block (system prompt, standing objective, hard constraints), and $L_a^{(k)}, L_o^{(k)}, L_r^{(k)}$ are the token counts of the action, raw observation, and reasoning trace at turn k.

3.2 Attention

For query position i and key position j within the same head and layer,

$$A_{i,j} = \frac{\exp(s_{i,j})}{\sum_{m=1}^{N_t} \exp(s_{i,m})}, \qquad s_{i,j} = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_k}}, \tag{2}$$

with $\mathbf{q}_i = \mathbf{X}_i W_Q$ and $\mathbf{k}_j = \mathbf{X}_j W_K$. We write $A_{i,\text{inv}} = \sum_{j \in \mathcal{S}_{\text{inv}}} A_{i,j}$ for the aggregate mass on the invariant block.

3.3 Partition

Fix t and partition the index set into

$$\mathcal{S}_{\text{inv}} : \text{invariant tokens}, \qquad |\mathcal{S}_{\text{inv}}| = L_u, \tag{3}$$

$$\mathcal{S}_{\text{noise}} : \text{raw observations}, \qquad |\mathcal{S}_{\text{noise}}| = \textstyle\sum_k L_o^{(k)}, \tag{4}$$

$$\mathcal{S}_{\text{work}} : \text{actions and reasoning}, \qquad |\mathcal{S}_{\text{work}}| = \textstyle\sum_k \big(L_a^{(k)} + L_r^{(k)}\big), \tag{5}$$

and write $N = |\mathcal{S}_{\text{noise}}|$ for the distractor count, treating $\mathcal{S}_{\text{work}}$ as part of the non-invariant pool where stated.

3.4 What is measured, and what is assumed

It is worth separating these two cleanly, because the empirical section measures one quantity and the theorems assume properties of another.

Measured: the attention distribution $A_{i,\cdot}$ for the final query position, read directly from the model in eager mode. No approximation, no surrogate embedding.

Assumed (only where stated): bounds on the projections, used solely for the worst-case result of §5. The threshold result of §4 requires no such assumption, which is one reason we prefer it.

4 The Dilution Threshold

This section contains the paper's most useful piece of mathematics, and it is short. The point is to replace a worst-case inequality over unknown norms with an identity over a single measurable quantity.

Proposition 1 (Two-group softmax identity)**.** *Fix a query i. Suppose the invariant keys share a common score s_{inv} and the non-invariant keys share a common score s_{non}, with sizes L_u and N respectively. Define the* effective logit gap

$$\Delta = s_{\text{inv}} - s_{\text{non}}. \tag{6}$$

Then

$$A_{i,\text{inv}} = \frac{L_u e^{s_{\text{inv}}}}{L_u e^{s_{\text{inv}}} + N e^{s_{\text{non}}}} = \frac{1}{1 + \dfrac{N}{L_u} e^{-\Delta}} = \sigma\left(\Delta + \ln \frac{L_u}{N}\right), \tag{7}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$.

Proof. Factor $e^{s_{\text{inv}}}$ from numerator and denominator and divide through by L_u. The logistic form follows from $1/(1+e^{-z}) = \sigma(z)$ with $z = \Delta + \ln(L_u/N)$, since $\dfrac{N}{L_u} e^{-\Delta} = e^{-\Delta - \ln(L_u/N)}$. □

Corollary 1 (Half-mass threshold)**.** *$A_{i,\text{inv}} > \frac{1}{2}$ if and only if*

$$\Delta > \ln \frac{N}{L_u} \qquad \Longleftrightarrow \qquad N < L_u e^{\Delta}. \tag{8}$$

Corollary 2 (Fractional guarantee under a cap)**.** *For any $\varepsilon \in (0,1)$, if $N \le C$ then*

$$A_{i,\text{inv}} \;\ge\; \frac{1}{1+\dfrac{C}{L_u}e^{-\Delta}} \;=\; \frac{L_u e^{\Delta}}{L_u e^{\Delta}+C}, \tag{9}$$

with equality when $N = C$.

Remark 1 (Why this is better than a norm bound). Proposition 1 contains no unknown constants. Equation (9) is an equality at $N = C$, so it is sharp for the two-group model rather than merely an upper or lower bound with slack. It requires no boundedness assumption on the projections. And every quantity in it is measurable: Δ is estimable from a handful of forward passes, as we do in §6. By contrast the norm-based bound of §5 depends on $\kappa = e^{2B}$, which for realistic projection norms exceeds 10^8, and which is uninformative at any practical context length.

Remark 2 (The logarithm is the whole phenomenon). Corollary 1 says that the invariant set holds the majority of attention exactly when the model's logit gap exceeds $\ln(N/L_u)$. The requirement is logarithmic in the distractor count. This has two consequences that pull in opposite directions, and both matter:

(i) **Compensation is cheap.** A model need only increase its invariant-versus-background gap by about $\ln N$ nats — for $N = 20{,}000$ and $L_u = 62$, roughly 5.8 nats — to hold the share constant. Since logits are unbounded and gaps of this size are well within the range trained models routinely produce, we should *expect* capable models to compensate, at least partially. §6 finds that the model we tested does exactly this.

(ii) **Dilution is not avoidable by scale.** Conversely, because the requirement grows with N without bound, any *fixed* logit gap Δ_0 is eventually overwhelmed: $N > L_u e^{\Delta_0}$ suffices, and no amount of training removes the dependence on N. A model that has learned a gap appropriate for 10^3 tokens has not thereby learned one appropriate for 10^7.

Remark 3 (The practical recipe). Corollary 2 inverts into a design rule. Given a target share $1-\varepsilon$ and an estimated gap Δ for the model in use, the mutable budget should satisfy

$$C \;\le\; L_u e^{\Delta}\left(\frac{1}{\varepsilon}-1\right). \tag{10}$$

For $\varepsilon = 0.5$ this is $C \le L_u e^{\Delta}$, the half-mass threshold. The formula is exact for the two-group model and requires only that the planner either measures Δ or chooses a conservative lower bound for it. This is the form in which the theory is actually usable, and §6 supplies the measurement for one model.

Remark 4 (Where the model is wrong). The two-group assumption is a simplification and we do not want to hide it. Real key sets have dispersed scores within each group, and Δ as defined above is the gap between group *means*, not a gap between actual logits. Three things follow. (i) Equation (7) is exact only for constant within-group scores; otherwise it defines an effective gap whose value is set by the observed share, which is how we estimate it below. (ii) The worst-case theorem of §5 is still available for the adversarial case, and it is the reason we keep it. (iii) Multi-head structure is ignored: $A_{i,\text{inv}}$ is defined per head, and a claim about the layer as a whole requires an aggregation convention, which we state when reporting measurements.

5 The Softmax Inundation Bound

We include the norm-based result for completeness and for the adversarial case, with its limitations attached.

Assumption 1 (Bounded projection energy)**.** There exist $M_Q, M_K > 0$ such that $\|\mathbf{q}_i\|_2 \le M_Q$ and $\|\mathbf{k}_j\|_2 \le M_K$ for all i, j. By Cauchy–Schwarz,

$$-B \le s_{i,j} \le B, \qquad B = \frac{M_Q M_K}{\sqrt{d_k}}, \tag{11}$$

and we write $\kappa = e^{2B}$.

Theorem 1 (Softmax Inundation Bound)**.** *Under Assumption 1, for every query i and turn t,*

$$A_{i,\mathrm{inv}} \;\le\; \frac{\kappa\, L_u}{L_u + N}, \qquad \textit{so} \qquad A_{i,\mathrm{inv}} = \mathcal{O}\!\left(\frac{L_u}{N}\right) \;\; \textit{as } N \to \infty. \tag{12}$$

Proof. Since $e^{-B} \le e^{s_{i,j}} \le e^{B}$ for all j,

$$\sum_{j \in \mathcal{S}_{\mathrm{inv}}} e^{s_{i,j}} \le L_u e^{B}, \tag{13}$$

and dropping the non-negative $\mathcal{S}_{\mathrm{work}}$ terms from the denominator of (2),

$$\sum_{m=1}^{N_t} e^{s_{i,m}} \ge (L_u + N) e^{-B}. \tag{14}$$

Dividing gives the claim. □

Remark 5 (This result is weak, and we say so). Equation (12) is vacuous whenever $\kappa L_u \gtrsim L_u + N$, since the right-hand side then exceeds 1 and the inequality carries no information. With $B = 10$, $\kappa \approx 5 \times 10^8$; the bound becomes informative only when N is within a factor of κ of L_u, which at practical scales it is not. The result is also loose in a second way: it bounds the share by assuming the *worst* scores for the invariant set and the *best* for the distractors, whereas the observed configuration is almost always the opposite. Its only genuine use is adversarial: it says that without any assumption on the score ordering, volume alone cannot promise a floor. Proposition 1 is the version to use when a gap estimate is available.

6 Empirical measurement

6.1 Method

We measure attention directly rather than modelling it. The probe (`experiments/live_attention_probe.py`) does the following:

(1) Loads `Qwen2.5-0.5B` in `float16` with `attn_implementation="eager"`, so that attention weights are materialized rather than fused away.

(2) Constructs a prompt with a fixed invariant prefix (62 tokens: a system role, a standing objective, and four hard constraints) followed by a variable number of distractor tokens drawn from a set of realistic log-line templates, followed by a fixed task question.

(3) Runs a single forward pass with `output_attentions=True` and reads the final query position's attention row over all keys.

(4) Reports the mean over all 24 layers and 14 query heads of the mass assigned to the invariant prefix.

Two methodological points deserve emphasis. First, we report the *final* query position because that is the position whose distribution governs the next generated token; averages over all positions would mix in early positions whose visible key set is shorter. Second, we validate the extraction against the stock implementation rather than trusting it: an earlier version of this probe computed the final attention row from hidden states via a hook, and we discarded that approach after it disagreed with eager attention by 1.4×10^{-2} in the maximum absolute deviation. The reported numbers come from the unmodified library path.

Remark 6 (What "invariant share" is not). It is not a measure of task success, and it is not a probability that the model will follow the instruction. It is the fraction of one head's attention mass at one position that lands on a fixed set of tokens. The reason to care is that a share near zero makes instruction-following close to impossible in that head, whereas a healthy share merely makes it possible.

6.2 Result 1: dilution as distractors grow

Distractors N	Total ctx.	Inv. share	Uniform	Enrichment
0	81	0.527	0.765	0.69×
64	146	0.427	0.425	1.00×
128	210	0.422	0.295	1.43×
256	338	0.405	0.183	2.21×
512	593	0.398	0.105	3.80×
1024	1106	0.381	0.056	6.80×
2048	2129	0.388	0.029	13.33×

Table 1: Qwen2.5-0.5B, final-query attention to a fixed 62-token invariant prefix, averaged over 24 layers and 14 heads. "Uniform" is the share that would obtain if attention were spread evenly, L_u/N_t. "Enrichment" is the ratio of the two. Distractor text is drawn from log-line templates; the task question is fixed.

Two features stand out, and only one of them was expected.

Expected: the invariant share declines as distractors accumulate, from 0.527 at $N = 0$ to 0.388 at $N = 2048$. With L_u fixed and N growing, the denominator of (2) grows and a fixed set of keys must lose share unless its relative scores grow.

Unexpected: the decline is very small. A 32× increase in distractor volume moves the share by about 9% relative. Simultaneously the enrichment over uniform grows from 1.0× to 13.3×: the model is attending to the prefix far more than chance, and increasingly so. The naive reading of (7) with a constant Δ would predict the share falling to roughly 0.01 by $N = 2048$; it does not.

6.3 Result 2: the effective gap grows logarithmically

We can make the discrepancy quantitative. Inverting (7) for the effective gap,

$$\hat{\Delta}(N) = \ln\left(\frac{L_u}{N}\left(\frac{1}{A_{i,\mathrm{inv}}} - 1\right)\right), \tag{15}$$

which is well defined whenever $A_{i,\mathrm{inv}} \in (0, 1)$.

Regressing $\hat{\Delta}$ on $\ln N$ gives

$$\hat{\Delta}(N) \;\approx\; 0.945 \ln N - 4.195, \qquad R^2 = 0.9996. \tag{16}$$

The slope is 0.945, and the theoretical boundary value is 1. A slope of exactly 1 would mean

$$\hat{\Delta}(N) = \ln N + c \quad \Longrightarrow \quad \frac{N}{L_u} e^{-\hat{\Delta}} = \frac{e^{-c}}{L_u} = \text{const}, \tag{17}$$

N	**Inv. share**	$\hat{\Delta}$ **(nats)**	$\ln(N/L_u)$	$L_u e^{\hat{\Delta}}$
64	0.427	−0.264	0.032	48
128	0.422	0.410	0.725	93
256	0.405	1.033	1.418	174
512	0.398	1.696	2.111	338
1024	0.381	2.321	2.804	631
2048	0.388	3.042	3.497	1299

Table 2: Effective logit gap recovered from the measured share via (15). The final column is the half-mass threshold $L_u e^{\Delta}$ from Corollary 1: below this N the invariant set holds the majority of attention.

so that by (7) the invariant share is *exactly constant* in N. The measured slope sits just below 1, which is why the observed share drifts slowly downward rather than being flat: the model compensates for context growth at 94.5% of the rate required to hold its allocation, and the residual 5.5% is what shows up as dilution.

Observation 1. On Qwen2.5-0.5B, over $N \in [64, 2048]$ and for this distractor distribution, the invariant attention share is an approximately scale-invariant function of distractor volume. The model supplies a logit gap that grows essentially as $\ln N$, matching the requirement identified in Remark 2 to within a small constant margin.

This is not, in our reading, a coincidence of a small model. Any training objective that rewards instruction-following across a range of context lengths will select for representations whose invariant-versus-background gap grows with length, because Remark 2 shows that such growth is available at logarithmic cost. Instruction tuning across varied sequence lengths is, in effect, a curriculum for exactly this property.

6.4 Result 3: capping stabilises what drift remains

If the model compensates almost perfectly, the residual drift is small, and the question becomes whether it is worth controlling. We test the turn-scale behaviour by simulating an append-only history against a capped one.

Turn	**Arm**	**Total ctx.**	**Inv. share**
2	append-only	202	0.423
2	bounded (C=240)	202	0.423
5	append-only	381	0.405
5	bounded	322	0.406
10	append-only	682	0.394
10	bounded	322	0.406
20	append-only	1282	0.388
20	bounded	322	0.406

Table 3: Turn-scaled trace, 60 distractor tokens per turn. The bounded arm holds mutable content at the cap, so total context plateaus and the share is constant thereafter. The append-only arm drifts down by about 8% relative over 20 turns.

Bounding the mutable tiers arrests the residual drift and, more importantly, makes it *provably* bounded rather than empirically small. That is the case for the guarantee in §9, and after Result 2 it is the honest case: not "we prevent collapse", but "we replace a property we observed with a property the controller enforces".

6.5 Threats to validity

We list these before the reader has to.

(a) **One model, one size.** All measurements are Qwen2.5-0.5B. Whether the 0.945 slope holds at 7B, 70B, or in models with different positional encodings is untested. The qualitative prediction from Remark 2 — slope at most 1, and higher for models trained at longer lengths — is testable and not tested.

(b) **Synthetic distractors.** The distractor text is drawn from eight log-line templates with a varying integer. Real tool output is far more heterogeneous. Homogeneity may make the group-mean gap (and hence $\hat{\Delta}$) unusually well defined; a distribution with a few very high-scoring distractors could behave differently, and Remark 4 of §5 names that case as adversarial and out of scope for a volume argument.

(c) **Attention is a proxy.** We measure allocation, not behaviour. The claim "allocation is preserved" does not entail "the instruction is followed". Correlating share with task success requires a behavioural benchmark, which we have not run.

(d) **Final-position attention only.** We read one row per head. This is the position that governs generation, but a model may route the relevant information through earlier positions or through the residual stream in ways a single row does not reveal.

(e) **GPU-limited reach.** The turn-scale experiment runs at 60 tokens per turn. The original configuration of 120 tokens per turn exceeded the 4 GB of available device memory in eager mode at 40 turns. Absolute numbers at realistic per-turn volumes will differ, though the qualitative plateau-versus-drift contrast should not.

(f) **Absence of collapse may be specific to the setting.** We do not observe inundation collapse, and we have not established the conditions under which it would occur. It is possible that a longer horizon than we can test on this hardware, or a distractor distribution with higher score variance, would produce it.

6.6 What the measurement actually settles

Claim	Status after measurement
Volume drives allocation through the softmax denominator	**Supported.** Share is monotone decreasing in N across the tested range, in the direction (7) requires.
A fixed logit gap is eventually overwhelmed by N	**Logically settled, not observed.** Follows from (7); the measured N range is too small to exhibit it.
Volume control holds the share fixed	**Supported.** Table 3: the capped arm is constant to three decimals.
Unbounded volume causes attention collapse in practice	**Not supported.** Share fell $\approx 9\%$ over a $32\times$ growth in N. The naive model over-predicts dilution substantially.
Volume control improves task success	**Untested.** Requires a behavioural benchmark.

Table 4: Status of the paper's claims after §6. We include the negative entries because they are the ones a reader is entitled to know.

7 Generational Context

7.1 Design

If §4 identifies volume as the controllable variable, the response is to control it explicitly. Generational Context applies the shape of generational garbage collection and of virtual-memory paging at the granularity of semantic context entries rather than KV blocks.

Definition 1 (Generational Context)**.** Active context is partitioned into three tiers:

Gen-1 *Tenured.* Founding objectives, system invariants, and hard constraints. Pinned; never evicted. Occupies L_u tokens.

Gen-2 *Evaporative.* Working memory: hypotheses, plans, uncommitted deltas, and committed-but-revisable state. Paged under a strict capacity C_1 by Attention-Weighted Eviction (§8).

Gen-3 *Ephemeral.* Raw tool output and logs. Passed through a distillation map σ_{distill} producing at most K_0 tokens, with the raw form discarded under a turn-based TTL.

Let C_{overhead} bound the structural tokens (role tags, delimiters, section headers) needed to render the window, and set

$$C_{\text{bound}} = C_1 + K_0 + C_{\text{overhead}}. \tag{18}$$

7.2 Why three tiers and not one budget

A single global budget would satisfy the volume bound of Theorem 2 equally well. The tiering earns its place on a different axis: it makes the eviction *policy* well posed. Under a single budget, a diagnostic observation and a standing constraint compete on the same footing, and attention-based scoring will sometimes evict the constraint, because in a given turn the constraint may genuinely be less attended than a salient error. Pinning removes that possibility for the tokens where it is intolerable, and the volume cap then bounds everything else.

The design is therefore: one tier that is exempt from competition, one tier that competes under a budget, and one tier that is summarized before it ever competes.

7.3 Interaction with the threshold result

Combining Definition 1 with Remark 2 gives a concrete sizing rule. With C_{bound} as the mutable budget, Corollary 2 becomes

$$A_{i,\text{inv}} \;\geq\; \frac{L_u e^{\Delta}}{L_u e^{\Delta} + C_{\text{bound}}}, \tag{19}$$

so a designer targeting $A_{i,\text{inv}} \geq 1 - \varepsilon$ chooses $C_{\text{bound}} \leq L_u e^{\Delta}(1/\varepsilon - 1)$. Nothing in this depends on the horizon, and nothing in it depends on the model continuing to compensate for context growth: the bound is maintained by the controller, not by the model's properties.

8 Attention-Weighted Eviction

8.1 Selection policy

The guarantee of §9 presumes a controller that maintains the volume ceiling. We implement eviction as an adaptation of ARC [4], which already solves recency-versus-frequency allocation for paged memory.

The cache maintains a recency list T_1, a frequency/saliency list T_2, ghost registries B_1, B_2 recording recently evicted entries, and an adaptation parameter p targeting the size of T_1. Victim selection within a list uses the *Information Density Score*

$$\mathrm{IDS}(E) = \frac{\mathrm{AttnWeight}(E) \cdot 100}{\mathrm{Entropy}(E) + 10^{-4}}, \tag{20}$$

with tier-based priors for entries whose attention statistics have not yet been measured. The intuition is that an entry is worth keeping when it receives attention predictably rather than diffusely: a high weight spread across a dispersed distribution is closer to noise than the same weight concentrated on a few tokens.

8.2 Post-condition

The invariant the controller enforces is

$$\sum_{e \in T_1 \cup T_2} \mathrm{tokens}(e) \;\le\; C \qquad \text{after every operation,} \tag{21}$$

with any single entry larger than C truncated before insertion. Equation (21) is a post-condition rather than a target: it holds after insertion, after promotion, and after distillation.

8.3 Two correctness traps

Both of these were found by testing rather than by inspection, and both are the kind of error that a bound like (21) would hide if it were assumed rather than checked.

(a) **Self-eviction on promotion.** Promoting an entry from T_1 to T_2 can grow the footprint, which triggers eviction, and a naive victim-selection rule can choose the newly promoted entry as its own victim. The entry is then silently lost in the same operation that promoted it, and because the operation reports success the loss is invisible. The fix is to exclude the incoming key from the victim pool: the routine must claim space from *other* entries or fail.

(b) **Unbounded ghost registries.** The ghost lists record recently evicted keys so that re-admission can be detected. If they are not themselves bounded, the bookkeeping grows without limit even though the active cache does not, which defeats the purpose of the controller while leaving (21) satisfied.

9 The Generational Retention Guarantee

Theorem 2 (General retention guarantee)**.** *Fix a query i and let Δ be the effective logit gap between the invariant set and the non-invariant set. If the active non-invariant volume is maintained at most C_{bound} at every turn, then for all $t \ge 1$,*

$$A_{i,\mathrm{inv}}^{(t)} \;\ge\; \frac{L_u e^{\Delta}}{L_u e^{\Delta} + C_{\mathrm{bound}}} \;>\; 0. \tag{22}$$

In particular the bound is independent of the horizon t.

Proof. By Definition 1 the non-invariant count satisfies $N_t \le C_1 + K_0 + C_{\mathrm{overhead}} = C_{\mathrm{bound}}$ at every turn, independently of t. Apply Corollary 2 with $C = C_{\mathrm{bound}}$. The right-hand side is positive because $L_u e^{\Delta} > 0$ and $C_{\mathrm{bound}} < \infty$. □

Corollary 3 (No horizon decay)**.** *Under the hypotheses of Theorem 2,* $\inf_{t \ge 1} A_{i,\mathrm{inv}}^{(t)} > 0$.

Remark 7 (Worst-case variant). The same conclusion holds without any assumption on the gap, at the cost of sharpness. Replacing Δ by the norm bound of Theorem 1 gives

$$A_{i,\mathrm{inv}}^{(t)} \;\geq\; \frac{L_u}{L_u + \kappa\, C_{\mathrm{bound}}}, \tag{23}$$

which is positive, bounded away from zero, and horizon-independent, but numerically uninformative at realistic κ. We state it only to record that the qualitative conclusion — no horizon decay — does not depend on the gap being favourable. The controller prevents *collapse*; the gap determines how much share is actually retained.

Remark 8 (What the guarantee is not). It is a statement about attention allocation under a maintained volume bound. It says that if the controller holds, then volume alone cannot drive the invariant share to zero. It does not say the share is large, that the model will follow the instruction, or that the task will succeed. An adversary who controls distractor *content* rather than count can violate the spirit of the result while leaving the bound intact, by constructing distractors whose scores dominate the invariant set within any fixed budget.

10 Intent auditing

The architecture includes a component for checking that a proposed action is consistent with the declared objective, and for leaving a verifiable trail of those checks. It is an engineering component, and we describe it as such.

An intent is serialized as a canonical digest over the declared goal, values, constraints, and priorities, and signed with Ed25519. A proposal is accepted only if its authorization token verifies against the current intent digest and its signatory holds the required role. Decisions are recorded in an append-only hash-chained log, so that a later dispute can be resolved by recomputation rather than by trusting the agent's account of what it decided.

Two honest caveats. The threat model is undocumented: we have not specified what the adversary is assumed to control, and without that the component should be read as a consistency mechanism rather than a security boundary. And the scenarios that exercise it are authored by the same team that wrote the checker, which is exactly the configuration in which a checker appears more capable than it is. Both gaps are addressable, and neither is addressed here.

11 Reference implementation

The architecture is implemented in the `generational_context` package and depends only on the Python standard library at runtime. It exposes tool observation, reasoning recording, and prompt compilation; it enforces the capacity post-condition (21) and the Gen-0 TTL behaviour.

The repository also contains:

- `ccb_bench.py`, the deterministic simulation harness reported in the companion paper [6];
- `experiments/live_attention_probe.py`, the measurement script behind §6, with its outputs committed as CSV;
- `tests/`, which exercises the tier invariants, the eviction post-condition including the self-eviction case, and the intent-audit primitives. The cryptographic tests skip rather than fail when the optional `cryptography` dependency is absent, matching the library's documented fallback.

None of this is a substitute for the behavioural experiment described in §12, and it is not offered as one.

12 Falsifiable predictions

A theory that cannot be wrong is not useful. The following are stated so they can be tested and, if the theory is mistaken, refuted.

P1. Slope bound. For any instruction-tuned model, the regression slope of $\hat{\Delta}$ on $\ln N$ is at most 1 in the regime where the invariant prefix is a plausible instruction block. A slope materially above 1 would mean the model *over*-compensates as context grows, which our account of the mechanism does not predict. *Test:* repeat §6 across model families and sizes.

P2. Slope scales with training length. Models trained or instruction-tuned at longer context lengths should exhibit slope closer to 1 than models tuned at shorter lengths, because longer training makes the logarithmic compensation more available. *Test:* regress slope against stated training context length across a model family.

P3. Distribution sensitivity. Replacing homogeneous log-line distractors with heterogeneous real tool traces should lower $\hat{\Delta}$ at fixed N, because a heterogeneous set produces higher maximum scores against the invariant block. *Test:* rerun the sweep with recorded agent traces in place of templates.

P4. Cap constant. In a capped arm the measured share should be constant in the number of turns to within measurement noise, and should track (9) as C varies. *Test:* sweep C and compare against the closed form.

P5. Behavioural coupling. Invariant share at the final position should correlate with instruction adherence in a multi-turn benchmark, weakly but positively. A null correlation would not refute the allocation mathematics, but it would remove the practical reason to care about it. *Test:* run a long-horizon agent benchmark with the controller on and off, reporting both share and success rate.

Prediction P5 is the one that decides whether this line of work matters. The mathematics in §§4–9 is, we believe, correct; the question of whether preserved allocation improves outcomes is empirical and open, and it is the experiment we most want to see run.

13 Limitations

We collect the limitations stated throughout, in one place.

(1) **The name is not a claim.** "Fool's Theorem" denotes a failure mode. The mathematical content of this paper is Proposition 1, Theorem 1, and Theorem 2, none of which is novel in technique. Proposition 1 is a rearrangement of the softmax definition; Theorem 1 is Cauchy–Schwarz.

(2) **The strongest result is conditional on a gap estimate.** The usable bound (9) needs Δ, which is model-, distribution-, layer-, and head-dependent. We measured it for one model on one distractor distribution. Treating it as a constant across settings would be a misuse.

(3) **The empirical evidence partly undercuts the motivation.** As Result 2 shows, the model we tested already compensates for context growth at roughly the required rate, and the dilution we measured is mild. The remedy is justified by making a property enforceable rather than by rescuing an observed collapse. Presenting it as the latter would misrepresent the data.

(4) **Single head, single model, synthetic distractors, single position.** See §6(a)–(d).

(5) **No behavioural evidence.** Nothing here shows that any of this improves task outcomes. It is the least interesting limitation to state and the most important.

(6) **Adversarial content is out of scope.** Volume control bounds count, not score. A bounded number of high-scoring distractors can dominate attention, and no result in this paper forbids it.

(7) **The intent-audit component is unvalidated** and has no documented threat model.

14 Related work

Attention sinks and streaming inference. Xiao et al. [9] observe that a small number of initial tokens absorb disproportionate attention, and exploit this to stream indefinitely. Our Observation 1 is consistent with a related effect: the prefix's advantage grows with context length rather than remaining fixed, which is what permits near-scale-invariant allocation. The two accounts differ in aim — theirs addresses degenerate KV statistics for continuity of generation, ours asks whether declared invariants keep competing — but they are likely describing overlapping machinery, and we would not be surprised if the sink mechanism is the substrate of the compensation we measure.

Position sensitivity. *Lost in the Middle* [3] documents that content in the interior of a long context is used less effectively than content at the edges; RULER [1] shows nominal window size overstates usable context. Both are compatible with our account and neither is subsumed by it: position and volume are distinct variables, and our measurement holds position fixed (the invariant prefix is always first) while varying volume.

KV-cache eviction. H2O [10] and SnapKV [2] select which KV entries to retain under a memory budget, using attention statistics as the salience signal. Attention-Weighted Eviction uses the same signal, and IDS (20) is a close relative of their scoring. The difference is granularity: they operate on KV entries within a forward pass to reduce memory, whereas we operate on semantic context entries across turns to bound a prompt. A reader who knows that literature should regard our eviction component as an application, not a contribution.

Agent memory. MemGPT [5] manages a virtual context window through function calls, with paging decisions made by the model itself. Generational Context makes those decisions with a deterministic controller, trading flexibility for predictable cost and for a capacity invariant that can be checked rather than hoped for. Pinning invariants corresponds to MemGPT's always-in-context core memory; the differences are the explicit tiering and the TTL treatment of raw observations.

Cache replacement. ARC [4] supplies the adaptation mechanism for recency versus frequency. We apply it unchanged in structure.

Attention. The underlying mechanism is the scaled dot-product attention of Vaswani et al. [8]; the rotary positional encoding used by the measured model is due to Su et al. [7].

15 Conclusion

We set out to explain a failure mode and ended up with a sharper account of its mechanism than we started with, and a weaker claim about its severity.

The mechanism is the softmax denominator, and it is best expressed as an identity rather than a bound: the invariant attention share equals $\sigma(\Delta + \ln(L_u/N))$, so the model must supply a logit gap that grows as $\ln N$ merely to hold its allocation constant. That logarithm is the whole phenomenon. It explains why compensation is cheap and therefore likely, and it explains why no fixed gap survives unbounded volume.

Measuring this on a real transformer produced the result we did not expect. The model's effective gap grows at a slope of 0.945 against $\ln N$, with $R^2 = 0.9996$ — almost exactly the rate required for scale-invariant allocation. Dilution is therefore far milder in practice than the uniform-score analysis suggests: about 9% over a 32× increase in distractor volume. We report this because it is the strongest evidence against our own framing, and because a reader who discovers it independently would be right to ask why we omitted it.

The systems contribution survives this, but it changes character. Generational Context does not rescue a collapsing share; it replaces a property we happened to observe with one the controller enforces. The retention guarantee is horizon-independent, it is stated in terms of a measurable gap, and it comes with an implementation whose capacity invariant is checked rather than assumed. Whether that enforced property buys anything in task outcomes is the open question, and §12 states what would settle it.

Acknowledgments

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